DSA + DAA Complete Revision Planner

60-Day Structured Roadmap • Topic-wise LeetCode • Pattern Questions • Advanced DAA

Duration: 60 Days **Topics: 13 DSA + 5 DAA** **LeetCode: 260+ Problems** **Patterns: 12 Types**

Category	Count	Coverage
DSA Topics	**13 Topics**	20 LeetCode questions each
Pattern Types	**12 Patterns**	10 questions per pattern
DAA Advanced Topics	**5 Topics**	10 questions each
60-Day Roadmap	**13 Phases**	Daily tasks + mock targets

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SECTION 1 — 60-Day DSA Roadmap

A structured, day-by-day plan to master DSA from fundamentals to advanced. Each phase includes concept revision, targeted LeetCode practice, and a timed mock session.

Period	Topic	Daily Tasks
Days 1–5	**Arrays, Hashing & Prefix Sums**	1. Revise array manipulation, prefix sum, difference array 2. Solve 4 LeetCode: Two Sum, Best Time to Buy, Maximum Subarray, Product Except Self 3. Study HashMap patterns: frequency count, running sum 4. Solve 4 more: Group Anagrams, Longest Consecutive, Top K Frequent, Subarray Sum = K 5. Mock: 2 random Easy + 1 Medium timed (45 min)
Days 6–10	**Two Pointers & Sliding Window**	1. Revise two-pointer template (sorted array, palindrome, three-sum) 2. Solve: Valid Palindrome, 3Sum, Container With Most Water, Trapping Rain Water 3. Revise sliding window (fixed vs variable), at-most-K trick 4. Solve: Longest Substring Without Repeat, Min Window Substring, Permutation in String 5. Mock: 1 Medium sliding window + 1 Medium two-pointer (60 min)
Days 11–15	**Stack, Queue & Monotonic Structures**	1. Revise stack (LIFO), queue (FIFO), deque patterns 2. Solve: Valid Parentheses, Min Stack, Daily Temperatures, Next Greater Element 3. Study monotonic stack: histogram, span problems 4. Solve: Largest Rectangle in Histogram, Car Fleet, Decode String, 132 Pattern 5. Mock: 2 Medium stack problems (45 min)
Days 16–20	**Binary Search**	1. Revise binary search templates (exact, lower/upper bound, rotated) 2. Solve: Binary Search, Search in Rotated Array, Find Min in Rotated Array 3. Study search-on-answer pattern 4. Solve: Koko Eating Bananas, Capacity to Ship, Split Array Largest Sum 5. Mock: 1 Hard binary search (60 min)
Days 21–25	**Linked List**	1. Revise: reversal, cycle detection (Floyd's), merge, two-pointer on list 2. Solve: Reverse Linked List, Merge Two Sorted Lists, Remove Nth Node, Linked List Cycle 3. Study: Copy with Random Pointer, Reorder List, LRU Cache 4. Solve: Merge K Sorted Lists, Reverse Nodes in k-Group, Add Two Numbers 5. Mock: 1 Medium + 1 Hard linked list (60 min)
Days 26–30	**Trees, BST & Tree DP**	1. Revise: DFS (pre/in/post-order), BFS level-order, BST properties 2. Solve: Invert Binary Tree, Max Depth, Same Tree, Level Order Traversal

Period	Topic	Daily Tasks
		3. Study: LCA, validate BST, diameter, path sum patterns 4. Solve: Binary Tree Max Path Sum, Serialize/Deserialize, Construct from Preorder+Inorder 5. Mock: 2 Medium + 1 Hard tree problem (75 min)
Days 31–35	**Heap & Priority Queue**	1. Revise: min-heap, max-heap, heapify, heap sort 2. Solve: Kth Largest in Stream, K Closest Points, Top K Frequent, Task Scheduler 3. Study: Find Median from Data Stream (two heaps), Merge K Sorted Lists 4. Solve: Design Twitter, IPO, Smallest Range from K Lists 5. Mock: 1 Medium heap + 1 Hard (60 min)
Days 36–40	**Graphs – BFS, DFS, Union-Find**	1. Revise: adjacency list/matrix, BFS for shortest path, DFS for components 2. Solve: Number of Islands, Clone Graph, Max Area of Island, Rotting Oranges 3. Study: Union-Find (path compression, rank), topological sort (Kahn's + DFS) 4. Solve: Course Schedule I & II, Redundant Connection, Accounts Merge 5. Mock: 1 Medium + 1 Hard graph (75 min)
Days 41–45	**Graphs – Shortest Path & Advanced**	1. Revise: Dijkstra's, Bellman-Ford, Floyd-Warshall 2. Solve: Network Delay Time, Cheapest Flights, Swim in Rising Water 3. Study: Minimum Spanning Tree (Kruskal/Prim), word ladder BFS 4. Solve: Min Cost to Connect Points, Word Ladder, Alien Dictionary 5. Mock: 1 Hard graph + review all graph patterns (90 min)
Days 46–52	**Dynamic Programming**	1. Revise: 1D DP (Fibonacci-style), 2D DP, state transitions 2. Solve: Climbing Stairs, House Robber, Coin Change, Unique Paths 3. Study: LCS, LIS (O(n log n)), partition DP 4. Solve: Longest Common Subsequence, Longest Increasing Subsequence, Partition Equal Subset Sum 5. Study: Interval DP, knapsack variants 6. Solve: Burst Balloons, Regular Expression Matching, Decode Ways, Interleaving String 7. Mock: 2 Medium + 1 Hard DP (90 min)
Days 53–56	**Backtracking & Trie**	1. Revise: choose-explore-unchoose template, pruning strategies 2. Solve: Subsets, Permutations, Combination Sum, N-Queens, Sudoku Solver 3. Revise: Trie node structure, insert/search/startsWith operations 4. Solve: Implement Trie, Word Search II, Design Add-Search Words, Search Suggestions
Days 57–58	**Greedy Algorithms**	1. Revise: greedy choice property, exchange argument proof technique 2. Solve: Jump Game I & II, Gas Station, Hand of Straights, Partition Labels

Period	Topic	Daily Tasks
		3. Solve: Non-overlapping Intervals, Meeting Rooms II, Task Scheduler, Merge Triplets
Days 59–60	**DAA Advanced & Mock Interviews**	1. Review DAA topics: Master Theorem, NP-completeness, approximation algorithms 2. Study: Matrix Chain Multiplication, Traveling Salesman (Bitmask DP), Convex Hull Trick 3. Full Mock Interview: 2 LeetCode problems in 90 min (1 Medium + 1 Hard) 4. Review all patterns, flashcard key complexities, final weak-area revision

SECTION 2 — Topic-wise LeetCode Questions

1. Arrays & Hashing

#	LC #	Problem Title (Clickable)	Difficulty
1	1	Two Sum	Easy
2	49	Group Anagrams	Medium
3	217	Contains Duplicate	Easy
4	238	Product of Array Except Self	Medium
5	36	Valid Sudoku	Medium
6	271	Encode and Decode Strings	Medium
7	128	Longest Consecutive Sequence	Medium
8	121	Best Time to Buy and Sell Stock	Easy
9	53	Maximum Subarray	Medium
10	1299	Replace Elements with Greatest Element on Right	Easy
11	283	Move Zeroes	Easy
12	75	Sort Colors	Medium
13	56	Merge Intervals	Medium
14	448	Find All Numbers Disappeared in Array	Easy
15	442	Find All Duplicates in Array	Medium
16	41	First Missing Positive	Hard
17	347	Top K Frequent Elements	Medium
18	242	Valid Anagram	Easy
19	560	Subarray Sum Equals K	Medium
20	334	Increasing Triplet Subsequence	Medium

2. Two Pointers

#	LC #	Problem Title (Clickable)	Difficulty
1	167	Two Sum II – Input Array Is Sorted	Medium
2	125	Valid Palindrome	Easy
3	15	3Sum	Medium
4	11	Container With Most Water	Medium

#	LC #	Problem Title (Clickable)	Difficulty
5	42	Trapping Rain Water	Hard
6	18	4Sum	Medium
7	392	Is Subsequence	Easy
8	680	Valid Palindrome II	Easy
9	283	Move Zeroes	Easy
10	905	Sort Array By Parity	Easy
11	259	3Sum Smaller	Medium
12	986	Interval List Intersections	Medium
13	1004	Max Consecutive Ones III	Medium
14	26	Remove Duplicates from Sorted Array	Easy
15	27	Remove Element	Easy
16	977	Squares of a Sorted Array	Easy
17	189	Rotate Array	Medium
18	844	Backspace String Compare	Easy
19	88	Merge Sorted Array	Easy
20	287	Find the Duplicate Number	Medium

3. Sliding Window

#	LC #	Problem Title (Clickable)	Difficulty
1	643	Maximum Average Subarray I	Easy
2	3	Longest Substring Without Repeating Characters	Medium
3	424	Longest Repeating Character Replacement	Medium
4	567	Permutation in String	Medium
5	76	Minimum Window Substring	Hard
6	239	Sliding Window Maximum	Hard
7	1004	Max Consecutive Ones III	Medium
8	1208	Get Equal Substrings Within Budget	Medium
9	209	Minimum Size Subarray Sum	Medium
10	438	Find All Anagrams in a String	Medium
11	1423	Maximum Points You Can Obtain from Cards	Medium
12	904	Fruit Into Baskets	Medium
13	930	Binary Subarrays With Sum	Medium
14	992	Subarrays with K Different Integers	Hard

#	LC #	Problem Title (Clickable)	Difficulty
15	1234	Replace the Substring for Balanced String	Medium
16	1100	Find K-Length Substrings With No Repeated Characters	Medium
17	1052	Grumpy Bookstore Owner	Medium
18	1695	Maximum Erasure Value	Medium
19	2024	Maximize the Confusion of an Exam	Medium
20	2090	K Radius Subarray Averages	Medium

4. Stack & Queue

#	LC #	Problem Title (Clickable)	Difficulty
1	20	Valid Parentheses	Easy
2	155	Min Stack	Medium
3	150	Evaluate Reverse Polish Notation	Medium
4	22	Generate Parentheses	Medium
5	739	Daily Temperatures	Medium
6	853	Car Fleet	Medium
7	84	Largest Rectangle in Histogram	Hard
8	496	Next Greater Element I	Easy
9	503	Next Greater Element II	Medium
10	901	Online Stock Span	Medium
11	225	Implement Stack using Queues	Easy
12	232	Implement Queue using Stacks	Easy
13	716	Max Stack	Hard
14	394	Decode String	Medium
15	735	Asteroid Collision	Medium
16	456	132 Pattern	Medium
17	316	Remove Duplicate Letters	Medium
18	402	Remove K Digits	Medium
19	321	Create Maximum Number	Hard
20	1190	Reverse Substrings Between Each Pair of Parentheses	Medium

5. Binary Search

#	LC #	Problem Title (Clickable)	Difficulty
1	704	Binary Search	Easy
2	74	Search a 2D Matrix	Medium
3	875	Koko Eating Bananas	Medium
4	33	Search in Rotated Sorted Array	Medium
5	153	Find Minimum in Rotated Sorted Array	Medium
6	981	Time Based Key-Value Store	Medium
7	4	Median of Two Sorted Arrays	Hard
8	34	Find First and Last Position of Element in Sorted Array	Medium
9	162	Find Peak Element	Medium
10	278	First Bad Version	Easy
11	410	Split Array Largest Sum	Hard
12	1011	Capacity To Ship Packages Within D Days	Medium
13	1231	Divide Chocolate	Hard
14	1283	Find the Smallest Divisor Given a Threshold	Medium
15	69	Sqrt(x)	Easy
16	374	Guess Number Higher or Lower	Easy
17	658	Find K Closest Elements	Medium
18	1482	Minimum Number of Days to Make m Bouquets	Medium
19	1870	Minimum Speed to Arrive on Time	Medium
20	2064	Minimized Maximum of Products Distributed to Any Store	Medium

6. Linked List

#	LC #	Problem Title (Clickable)	Difficulty
1	206	Reverse Linked List	Easy
2	21	Merge Two Sorted Lists	Easy
3	143	Reorder List	Medium
4	19	Remove Nth Node From End of List	Medium
5	138	Copy List with Random Pointer	Medium
6	2	Add Two Numbers	Medium
7	141	Linked List Cycle	Easy
8	287	Find the Duplicate Number	Medium
9	146	LRU Cache	Medium
10	23	Merge K Sorted Lists	Hard

#	LC #	Problem Title (Clickable)	Difficulty
11	25	Reverse Nodes in k-Group	Hard
12	92	Reverse Linked List II	Medium
13	82	Remove Duplicates from Sorted List II	Medium
14	160	Intersection of Two Linked Lists	Easy
15	234	Palindrome Linked List	Easy
16	148	Sort List	Medium
17	328	Odd Even Linked List	Medium
18	430	Flatten a Multilevel Doubly Linked List	Medium
19	708	Insert into a Sorted Circular Linked List	Medium
20	432	All O(1) Data Structure	Hard

7. Trees & BST

#	LC #	Problem Title (Clickable)	Difficulty
1	226	Invert Binary Tree	Easy
2	104	Maximum Depth of Binary Tree	Easy
3	543	Diameter of Binary Tree	Easy
4	110	Balanced Binary Tree	Easy
5	100	Same Tree	Easy
6	572	Subtree of Another Tree	Easy
7	235	Lowest Common Ancestor of a BST	Medium
8	102	Binary Tree Level Order Traversal	Medium
9	199	Binary Tree Right Side View	Medium
10	1448	Count Good Nodes in Binary Tree	Medium
11	98	Validate Binary Search Tree	Medium
12	230	Kth Smallest Element in a BST	Medium
13	105	Construct Binary Tree from Preorder and Inorder Traversal	Medium
14	124	Binary Tree Maximum Path Sum	Hard
15	297	Serialize and Deserialize Binary Tree	Hard
16	236	Lowest Common Ancestor of a Binary Tree	Medium
17	450	Delete Node in a BST	Medium
18	437	Path Sum III	Medium
19	112	Path Sum	Easy
20	662	Maximum Width of Binary Tree	Medium

8. Heap & Priority Queue

#	LC #	Problem Title (Clickable)	Difficulty
1	703	Kth Largest Element in a Stream	Easy
2	1046	Last Stone Weight	Easy
3	973	K Closest Points to Origin	Medium
4	215	Kth Largest Element in an Array	Medium
5	621	Task Scheduler	Medium
6	355	Design Twitter	Medium
7	295	Find Median from Data Stream	Hard
8	378	Kth Smallest Element in a Sorted Matrix	Medium
9	347	Top K Frequent Elements	Medium
10	23	Merge K Sorted Lists	Hard
11	502	IPO	Hard
12	632	Smallest Range Covering Elements from K Lists	Hard
13	1675	Minimize Deviation in Array	Hard
14	1882	Process Tasks Using Servers	Medium
15	871	Minimum Number of Refueling Stops	Hard
16	407	Trapping Rain Water II	Hard
17	480	Sliding Window Median	Hard
18	218	The Skyline Problem	Hard
19	2402	Meeting Rooms III	Hard
20	2542	Maximum Subsequence Score	Medium

9. Graphs

#	LC #	Problem Title (Clickable)	Difficulty
1	200	Number of Islands	Medium
2	133	Clone Graph	Medium
3	695	Max Area of Island	Medium
4	417	Pacific Atlantic Water Flow	Medium
5	130	Surrounded Regions	Medium
6	994	Rotting Oranges	Medium
7	286	Walls and Gates	Medium
8	207	Course Schedule	Medium

#	LC #	Problem Title (Clickable)	Difficulty
9	210	Course Schedule II	Medium
10	684	Redundant Connection	Medium
11	323	Number of Connected Components in an Undirected Graph	Medium
12	261	Graph Valid Tree	Medium
13	127	Word Ladder	Hard
14	269	Alien Dictionary	Hard
15	743	Network Delay Time	Medium
16	787	Cheapest Flights Within K Stops	Medium
17	1091	Shortest Path in Binary Matrix	Medium
18	778	Swim in Rising Water	Hard
19	1584	Min Cost to Connect All Points	Medium
20	332	Reconstruct Itinerary	Hard

10. Dynamic Programming

#	LC #	Problem Title (Clickable)	Difficulty
1	70	Climbing Stairs	Easy
2	746	Min Cost Climbing Stairs	Easy
3	198	House Robber	Medium
4	213	House Robber II	Medium
5	5	Longest Palindromic Substring	Medium
6	647	Palindromic Substrings	Medium
7	91	Decode Ways	Medium
8	322	Coin Change	Medium
9	152	Maximum Product Subarray	Medium
10	139	Word Break	Medium
11	300	Longest Increasing Subsequence	Medium
12	416	Partition Equal Subset Sum	Medium
13	62	Unique Paths	Medium
14	1143	Longest Common Subsequence	Medium
15	309	Best Time to Buy and Sell Stock with Cooldown	Medium
16	518	Coin Change II	Medium
17	494	Target Sum	Medium
18	97	Interleaving String	Medium

#	LC #	Problem Title (Clickable)	Difficulty
19	312	Burst Balloons	Hard
20	10	Regular Expression Matching	Hard

11. Backtracking

#	LC #	Problem Title (Clickable)	Difficulty
1	78	Subsets	Medium
2	90	Subsets II	Medium
3	46	Permutations	Medium
4	47	Permutations II	Medium
5	39	Combination Sum	Medium
6	40	Combination Sum II	Medium
7	22	Generate Parentheses	Medium
8	79	Word Search	Medium
9	131	Palindrome Partitioning	Medium
10	51	N-Queens	Hard
11	52	N-Queens II	Hard
12	37	Sudoku Solver	Hard
13	212	Word Search II	Hard
14	93	Restore IP Addresses	Medium
15	17	Letter Combinations of a Phone Number	Medium
16	77	Combinations	Medium
17	216	Combination Sum III	Medium
18	89	Gray Code	Medium
19	526	Beautiful Arrangement	Medium
20	291	Word Pattern II	Medium

12. Greedy

#	LC #	Problem Title (Clickable)	Difficulty
1	53	Maximum Subarray	Medium
2	55	Jump Game	Medium
3	45	Jump Game II	Medium

#	LC #	Problem Title (Clickable)	Difficulty
4	134	Gas Station	Medium
5	846	Hand of Straights	Medium
6	1899	Merge Triplets to Form Target Triplet	Medium
7	763	Partition Labels	Medium
8	678	Valid Parenthesis String	Medium
9	1029	Two City Scheduling	Medium
10	406	Queue Reconstruction by Height	Medium
11	435	Non-overlapping Intervals	Medium
12	452	Minimum Number of Arrows to Burst Balloons	Medium
13	605	Can Place Flowers	Easy
14	714	Best Time to Buy and Sell Stock with Transaction Fee	Medium
15	621	Task Scheduler	Medium
16	1353	Maximum Number of Events That Can Be Attended	Medium
17	1642	Furthest Building You Can Reach	Medium
18	1710	Maximum Units on a Truck	Easy
19	1717	Maximum Score From Removing Substrings	Medium
20	2311	Longest Binary Subsequence Less Than or Equal to K	Medium

13. Trie

#	LC #	Problem Title (Clickable)	Difficulty
1	208	Implement Trie (Prefix Tree)	Medium
2	211	Design Add and Search Words Data Structure	Medium
3	212	Word Search II	Hard
4	648	Replace Words	Medium
5	677	Map Sum Pairs	Medium
6	1268	Search Suggestions System	Medium
7	421	Maximum XOR of Two Numbers in an Array	Medium
8	720	Longest Word in Dictionary	Medium
9	336	Palindrome Pairs	Hard
10	745	Prefix and Suffix Search	Hard
11	1698	Number of Distinct Substrings in a String	Medium
12	820	Short Encoding of Words	Medium
13	1803	Count Pairs With XOR in a Range	Hard

#	LC #	Problem Title (Clickable)	Difficulty
14	2227	Encrypt and Decrypt Strings	Hard
15	1707	Maximum XOR With an Element From Array	Hard
16	676	Implement Magic Dictionary	Medium
17	1166	Design File System	Medium
18	1858	Longest Word With All Prefixes	Medium
19	2416	Sum of Prefix Scores of Strings	Hard
20	1032	Stream of Characters	Hard

SECTION 3 — Pattern-wise Questions

Pattern 1: Fast & Slow Pointers (Floyd's Cycle)

#	LC #	Problem Title (Clickable)	Difficulty
1	141	Linked List Cycle	Easy
2	142	Linked List Cycle II	Medium
3	287	Find the Duplicate Number	Medium
4	202	Happy Number	Easy
5	876	Middle of the Linked List	Easy
6	234	Palindrome Linked List	Easy
7	457	Circular Array Loop	Medium
8	143	Reorder List	Medium
9	148	Sort List	Medium
10	708	Insert into a Sorted Circular Linked List	Medium

Pattern 2: Merge Intervals

#	LC #	Problem Title (Clickable)	Difficulty
1	56	Merge Intervals	Medium
2	57	Insert Interval	Medium
3	435	Non-overlapping Intervals	Medium
4	252	Meeting Rooms	Easy
5	253	Meeting Rooms II	Medium
6	452	Minimum Number of Arrows to Burst Balloons	Medium
7	986	Interval List Intersections	Medium
8	1288	Remove Covered Intervals	Medium
9	759	Employee Free Time	Hard
10	715	Range Module	Hard

Pattern 3: Monotonic Stack / Queue

#	LC #	Problem Title (Clickable)	Difficulty
1	739	Daily Temperatures	Medium
2	496	Next Greater Element I	Easy
3	503	Next Greater Element II	Medium
4	84	Largest Rectangle in Histogram	Hard
5	85	Maximal Rectangle	Hard
6	901	Online Stock Span	Medium
7	456	132 Pattern	Medium
8	907	Sum of Subarray Minimums	Medium
9	402	Remove K Digits	Medium
10	239	Sliding Window Maximum	Hard

Pattern 4: Union Find (Disjoint Sets)

#	LC #	Problem Title (Clickable)	Difficulty
1	547	Number of Provinces	Medium
2	684	Redundant Connection	Medium
3	685	Redundant Connection II	Hard
4	721	Accounts Merge	Medium
5	128	Longest Consecutive Sequence	Medium
6	200	Number of Islands	Medium
7	323	Number of Connected Components	Medium
8	1319	Number of Operations to Make Network Connected	Medium
9	1202	Smallest String With Swaps	Medium
10	959	Regions Cut by Slashes	Medium

Pattern 5: Topological Sort

#	LC #	Problem Title (Clickable)	Difficulty
1	207	Course Schedule	Medium
2	210	Course Schedule II	Medium
3	269	Alien Dictionary	Hard
4	310	Minimum Height Trees	Medium
5	329	Longest Increasing Path in a Matrix	Hard

#	LC #	Problem Title (Clickable)	Difficulty
6	444	Sequence Reconstruction	Medium
7	802	Find Eventual Safe States	Medium
8	851	Loud and Rich	Medium
9	1203	Sort Items by Groups Respecting Dependencies	Hard
10	2115	Find All Possible Recipes from Given Supplies	Medium

Pattern 6: Binary Search on Answer

#	LC #	Problem Title (Clickable)	Difficulty
1	875	Koko Eating Bananas	Medium
2	1011	Capacity To Ship Packages Within D Days	Medium
3	410	Split Array Largest Sum	Hard
4	1231	Divide Chocolate	Hard
5	1482	Minimum Number of Days to Make m Bouquets	Medium
6	1283	Find the Smallest Divisor Given a Threshold	Medium
7	1870	Minimum Speed to Arrive on Time	Medium
8	2064	Minimized Maximum of Products Distributed	Medium
9	774	Minimize Max Distance to Gas Station	Hard
10	1760	Minimum Limit of Balls in a Bag	Medium

Pattern 7: Tree BFS / Level Order

#	LC #	Problem Title (Clickable)	Difficulty
1	102	Binary Tree Level Order Traversal	Medium
2	107	Binary Tree Level Order Traversal II	Medium
3	103	Binary Tree Zigzag Level Order Traversal	Medium
4	199	Binary Tree Right Side View	Medium
5	637	Average of Levels in Binary Tree	Easy
6	515	Find Largest Value in Each Tree Row	Medium
7	116	Populating Next Right Pointers in Each Node	Medium
8	994	Rotting Oranges	Medium
9	542	01 Matrix	Medium
10	1162	As Far from Land as Possible	Medium

Pattern 8: Tree DFS / Recursion

#	LC #	Problem Title (Clickable)	Difficulty
1	112	Path Sum	Easy
2	113	Path Sum II	Medium
3	437	Path Sum III	Medium
4	124	Binary Tree Maximum Path Sum	Hard
5	543	Diameter of Binary Tree	Easy
6	297	Serialize and Deserialize Binary Tree	Hard
7	105	Construct Binary Tree from Preorder and Inorder	Medium
8	110	Balanced Binary Tree	Easy
9	114	Flatten Binary Tree to Linked List	Medium
10	98	Validate Binary Search Tree	Medium

Pattern 9: Knapsack DP Pattern

#	LC #	Problem Title (Clickable)	Difficulty
1	416	Partition Equal Subset Sum	Medium
2	494	Target Sum	Medium
3	474	Ones and Zeroes	Medium
4	1049	Last Stone Weight II	Medium
5	518	Coin Change II	Medium
6	322	Coin Change	Medium
7	377	Combination Sum IV	Medium
8	879	Profitable Schemes	Hard
9	1155	Number of Dice Rolls With Target Sum	Medium
10	2787	Ways to Express an Integer as Sum of Powers	Medium

Pattern 10: Interval DP

#	LC #	Problem Title (Clickable)	Difficulty
1	5	Longest Palindromic Substring	Medium
2	516	Longest Palindromic Subsequence	Medium
3	312	Burst Balloons	Hard

#	LC #	Problem Title (Clickable)	Difficulty
4	1039	Minimum Score Triangulation of Polygon	Medium
5	664	Strange Printer	Hard
6	1547	Minimum Cost to Cut a Stick	Hard
7	1000	Minimum Cost to Merge Stones	Hard
8	96	Unique Binary Search Trees	Medium
9	10	Regular Expression Matching	Hard
10	44	Wildcard Matching	Hard

Pattern 11: Shortest Path (Dijkstra / BFS)

#	LC #	Problem Title (Clickable)	Difficulty
1	743	Network Delay Time	Medium
2	787	Cheapest Flights Within K Stops	Medium
3	1091	Shortest Path in Binary Matrix	Medium
4	778	Swim in Rising Water	Hard
5	1514	Path with Maximum Probability	Medium
6	1631	Path With Minimum Effort	Medium
7	1334	Find the City With the Smallest Number of Neighbors	Medium
8	882	Reachable Nodes in Subdivided Graph	Hard
9	1368	Minimum Cost to Make at Least One Valid Path in a Grid	Hard
10	2290	Minimum Obstacle Removal to Reach Corner	Hard

Pattern 12: Matrix / Grid BFS-DFS

#	LC #	Problem Title (Clickable)	Difficulty
1	200	Number of Islands	Medium
2	695	Max Area of Island	Medium
3	130	Surrounded Regions	Medium
4	417	Pacific Atlantic Water Flow	Medium
5	329	Longest Increasing Path in a Matrix	Hard
6	827	Making A Large Island	Hard
7	1020	Number of Enclaves	Medium
8	934	Shortest Bridge	Medium

#	LC #	Problem Title (Clickable)	Difficulty
9	**980**	Unique Paths III	**Hard**
10	**1905**	Count Sub Islands	**Medium**

SECTION 4 — Revision Notes & Key Concepts

Arrays & Hashing

☐ **Resources:** NeetCode Roadmap | ► YouTube Tutorial | CP-Algorithms Ref

Key Concept / Pattern / Formula
1. Prefix Sum: preSum[i] = preSum[i-1] + arr[i]. Subarray sum [l,r] = preSum[r] - preSum[l-1].
2. HashMap counting: Use map to store frequency / first-occurrence index.
3. Kadane's Algorithm: maxSub = max(nums[i], maxSub + nums[i]); globalMax = max(globalMax, maxSub).
4. Sort + Two Pointer for pair/triplet problems to reduce from $O(n^2) \rightarrow O(n \log n)$.
5. Modular Hashing: (sum % k == 0) ↔ two indices with same running_sum % k.

Two Pointers

☐ **Resources:** NeetCode Roadmap | ► YouTube Tutorial | CP-Algorithms Ref

Key Concept / Pattern / Formula
1. Left/Right shrink: Works only on SORTED arrays. Move pointer that doesn't satisfy condition.
2. Fast/Slow: Slow moves 1 step, fast 2. They meet at cycle entry (Floyd's proof).
3. Three-sum: Fix one element, run two-pointer on rest. Skip duplicates explicitly.
4. Palindrome check: Move both pointers inward; break on mismatch.
5. Partition: Dutch National Flag — three-way partition in O(n) time.

Sliding Window

☐ **Resources:** NeetCode Roadmap | ► YouTube Tutorial | CP-Algorithms Ref

Key Concept / Pattern / Formula
1. Fixed window: Slide by adding right element and removing left element each step.
2. Variable window: Expand right; shrink left while window is invalid.
3. At-most-K trick: atMost(k) - atMost(k-1) gives exactly-K subarrays count.
4. Frequency map: Use a counter of chars; track how many are 'satisfied'.

Key Concept / Pattern / Formula
5. Deque for max/min: Maintain monotonic deque for O(1) window max queries.

Binary Search

☐ **Resources:** NeetCode Roadmap | ▶ YouTube Tutorial | CP-Algorithms Ref

Key Concept / Pattern / Formula
1. Template: lo=0, hi=n-1; while lo<=hi: mid=(lo+hi)//2; update lo/hi based on condition.
2. Search on answer: Binary search over the answer space when you can verify feasibility.
3. Rotated array: One half is always sorted. Determine which, then standard search.
4. Lower bound: First index where arr[i] >= target. Upper bound: first index where arr[i] > target.
5. Overflow-safe mid: mid = lo + (hi - lo) / 2.

Stack & Queue

☐ **Resources:** NeetCode Roadmap | ▶ YouTube Tutorial | CP-Algorithms Ref

Key Concept / Pattern / Formula
1. Monotonic Stack: Maintain increasing or decreasing order; pop when current violates order.
2. Next Greater Element: Push indices; pop when nums[stack.top] < nums[i].
3. Balanced Parentheses: Push opens; on close check top matches.
4. Min Stack: Maintain separate min-stack or store (val, currentMin) pairs.
5. Deque as queue: popleft() for BFS. Use collections.deque for O(1) operations.

Linked List

☐ **Resources:** NeetCode Roadmap | ▶ YouTube Tutorial | CP-Algorithms Ref

Key Concept / Pattern / Formula
1. Reverse in-place: prev=None, curr=head; while curr: next=curr.next; curr.next=prev; prev=curr; curr=next.
2. Find middle: Slow/fast pointers — when fast reaches end, slow is at middle.
3. Merge two sorted: Use dummy head node to simplify edge cases.
4. Detect cycle: Floyd's — they meet iff cycle exists; then reset one to head, step both by 1.

Key Concept / Pattern / Formula
5. Dummy node: Always use a dummy predecessor to handle head deletion cleanly.

Trees & BST

☐ **Resources:** NeetCode Roadmap | ► YouTube Tutorial | CP-Algorithms Ref

Key Concept / Pattern / Formula
1. In-order of BST = sorted array. Use this for kth-smallest, validation, etc.
2. DFS returns: Often return (result, subtree_height) or similar aggregate.
3. LCA: If both nodes in left subtree, go left. If both right, go right. Else current node is LCA.
4. Serialize: Preorder with null markers. Deserialize with queue/index pointer.
5. Balanced check: Return -1 as sentinel for 'unbalanced' flag up the recursion.

Heap & Priority Queue

☐ **Resources:** NeetCode Roadmap | ► YouTube Tutorial | CP-Algorithms Ref

Key Concept / Pattern / Formula
1. Min-heap for Kth largest: maintain heap of size K; root is answer.
2. Max-heap for Kth smallest: Use negative values in Python's heapq (min-heap only).
3. Merge K sorted: Push (val, list_idx, elem_idx) triples into heap.
4. Running median: Two heaps (max-heap for lower half, min-heap for upper half). Balance sizes.
5. Heap sort: Build heap O(n), then extract-min n times O(n log n).

Graphs

☐ **Resources:** NeetCode Roadmap | ► YouTube Tutorial | CP-Algorithms Ref

Key Concept / Pattern / Formula
1. BFS: Queue-based. Shortest path in unweighted graph. Mark visited before enqueue.
2. DFS: Stack/recursion. Cycle detection, connected components, topological sort.
3. Dijkstra: Priority queue with (dist, node). Relax neighbors. O((V+E) log V).
4. Bellman-Ford: Relax all edges V-1 times. Handles negative weights. O(VE).

Key Concept / Pattern / Formula
5. Union-Find: path compression + union by rank → near O(1) per operation amortized.

Dynamic Programming

☐ **Resources:** NeetCode Roadmap | ▶ YouTube Tutorial | CP-Algorithms Ref

Key Concept / Pattern / Formula
1. Identify: Overlapping subproblems + optimal substructure = DP.
2. 1D DP: dp[i] depends on dp[i-1] (or a window). Often reducible to O(1) space.
3. 2D DP: dp[i][j] = f(dp[i-1][j], dp[i][j-1], dp[i-1][j-1]).
4. LIS: O(n log n) with patience sorting / binary search on 'tails' array.
5. Memoization vs Tabulation: Top-down with @cache for clarity; bottom-up for space optimization.

Backtracking

☐ **Resources:** NeetCode Roadmap | ▶ YouTube Tutorial | CP-Algorithms Ref

Key Concept / Pattern / Formula
1. Template: choose → recurse → unchoose (restore state).
2. Pruning: Add conditions before recursing to skip impossible branches early.
3. Avoid duplicates: Sort input; skip if nums[i] == nums[i-1] at the same recursion level.
4. Permutations vs Combinations: Permutations use 'used[]' array; combinations use start index.
5. State space: Each recursion level = one decision point. Depth = length of solution.

SECTION 5 — DAA Advanced Questions

These questions go beyond standard LeetCode and test deep algorithmic thinking: correctness proofs, complexity analysis, reductions, and optimization techniques required for competitive programming, research, and top-tier SDE interviews.

1. Divide & Conquer

#	DAA Question / Topic	Reference
1	Merge Sort – implement and analyze T(n) = 2T(n/2) + O(n)	CP-Algo
2	Quick Sort – randomized pivot, worst/average case analysis	CP-Algo
3	Closest Pair of Points – O(n log n) geometric D&C	Wikipedia
4	Strassen's Matrix Multiplication – T(n) = 7T(n/2) + $O(n^2)$	Wikipedia
5	Maximum Subarray (D&C version) – cross-subarray merge step	CP-Algo
6	Count Inversions in Array – modified merge sort	Wikipedia
7	Order Statistics (Median of Medians) – guaranteed O(n) selection	LeetCode
8	Karatsuba Multiplication – O(n^1.585) integer multiplication	CP-Algo
9	Fast Fourier Transform (FFT) – polynomial multiplication in O(n log n)	Wikipedia
10	Binary Exponentiation (Fast Power) – O(log n) modular exponentiation	Wikipedia

2. Greedy Algorithms

#	DAA Question / Topic	Reference
1	Activity Selection Problem – proof of greedy optimality	CP-Algo
2	Huffman Coding – optimal prefix-free encoding, prove by exchange argument	CP-Algo
3	Fractional Knapsack – why greedy fails for 0/1 knapsack	Wikipedia
4	Dijkstra's Algorithm – greedy proof via invariant	CP-Algo
5	Prim's & Kruskal's MST – cut property and cycle property proofs	CP-Algo
6	Job Scheduling with Deadlines – maximize profit greedy	CP-Algo
7	Egyptian Fraction – greedy decomposition, termination proof	Wikipedia
8	Optimal Merge Pattern – minimize total merge cost using min-heap	Wikipedia
9	Interval Graph Coloring – minimum number of classrooms problem	Wikipedia
10	Set Cover Approximation – greedy gives O(ln n) approximation	Wikipedia

3. Dynamic Programming (Advanced)

#	DAA Question / Topic	Reference
1	Matrix Chain Multiplication – $O(n^3)$ DP, optimal parenthesization	□ CP-Algo
2	Optimal BST (Knuth's optimization) – $O(n^2)$ with divide-and-conquer speedup	□ CP-Algo
3	Edit Distance (Levenshtein) – 3 operations, space optimization	□ LeetCode
4	0/1 Knapsack – pseudo-polynomial, FPTAS approximation scheme	□ Wikipedia
5	Traveling Salesman Problem (Bitmask DP) – $O(n^2 \cdot 2^n)$ Held-Karp	□ CP-Algo
6	Longest Common Subsequence – LCS reconstruction and variants	□ CP-Algo
7	DP on Trees – subtree DP, rerooting technique	□ CF Blog
8	Digit DP – count integers in [L, R] satisfying property	□ CF Blog
9	Bitmasked DP for assignment problems	□ CF Blog
10	Convex Hull Trick / Li Chao Tree – $O(n \log n)$ DP optimization	□ CP-Algo

4. Graph Algorithms

#	DAA Question / Topic	Reference
1	Strongly Connected Components – Tarjan's / Kosaraju's algorithm	□ CP-Algo
2	Articulation Points & Bridges – low-link values in DFS	□ CP-Algo
3	Max Flow Min Cut – Ford-Fulkerson, prove max-flow = min-cut	□ CP-Algo
4	Bipartite Matching (Hopcroft-Karp) – $O(\sqrt{V} \cdot E)$	□ CP-Algo
5	Floyd-Warshall APSP – $O(V^3)$, negative cycle detection	□ CP-Algo
6	Johnson's Algorithm – APSP with negative edges $O(V^2 \log V + VE)$	□ CP-Algo
7	Euler Circuit / Hamiltonian Path – existence conditions, complexity gap	□ Wikipedia
8	2-SAT Problem – SCC-based linear time solution	□ CP-Algo
9	Planar Graph – Euler's formula, planarity testing	□ Wikipedia
10	Graph Coloring – chromatic polynomial, NP-hardness proof	□ Wikipedia

5. NP Theory & Complexity

#	DAA Question / Topic	Reference
1	P vs NP – formal definitions, implications of P = NP	□ Wikipedia
2	Polynomial-time reduction – show 3-SAT $\leq_p$ CLIQUE	□ Wikipedia
3	NP-Completeness of Vertex Cover – reduction from Independent Set	□ Wikipedia

#	DAA Question / Topic	Reference
4	Subset Sum is NP-Complete – reduction from 3-SAT	□ Wikipedia
5	Approximation algorithms – vertex cover 2-approx, TSP 1.5-approx (Christofides)	□ Wikipedia
6	Randomized algorithms – Las Vegas vs Monte Carlo, probability amplification	□ Wikipedia
7	Amortized analysis – aggregate, accounting, potential method	□ Wikipedia
8	Space complexity classes – PSPACE, NPSPACE, Savitch's theorem	□ Wikipedia
9	Parameterized complexity – FPT, W[1]-hard, kernelization	□ Wikipedia
10	Online algorithms – competitive analysis, ski rental problem	□ Wikipedia

Complexity Quick Reference

Algorithm / Operation	Time	Space
Binary Search	$O(\log n)$	$O(1)$
Merge Sort	$O(n \log n)$	$O(n)$
Quick Sort (avg)	$O(n \log n)$	$O(\log n)$
Heap Insert/Extract	$O(\log n)$	$O(n)$
BFS / DFS (graph)	$O(V + E)$	$O(V)$
Dijkstra (binary heap)	$O((V+E) \log V)$	$O(V)$
Floyd-Warshall	$O(V^3)$	$O(V^2)$
Topological Sort	$O(V + E)$	$O(V)$
DP – LCS	$O(mn)$	$O(mn) \to O(n)$
DP – Knapsack	$O(nW)$	$O(nW) \to O(W)$
DP – Matrix Chain	$O(n^3)$	$O(n^2)$
Trie Insert/Search	$O(L)$	$O(L \cdot \Sigma)$
Union-Find (opt.)	$O(\alpha(n)) \approx O(1)$	$O(n)$
KMP String Match	$O(n + m)$	$O(m)$

FINAL TIPS FOR SUCCESS

- Always understand WHY a solution works before moving on — write the intuition in your own words.
- For every problem, identify the pattern first: sliding window? DP? graph traversal? Then code.
- Time yourself: Easy ≤ 15 min, Medium ≤ 30 min, Hard ≤ 60 min. Practice under pressure.
- Review wrong answers within 24 hours and add them to a personal 'mistake journal'.
- For DP problems: define state → write recurrence → code top-down → optimize to bottom-up.
- For graph problems: draw the graph on paper first. Edge cases: disconnected, self-loops, directed vs undirected.
- Mocks matter: do at least 2 full mock interviews per week in the last 3 weeks.
- Master complexities: know time AND space for every algorithm. Interviewers ask both.
- Read editorial after solving to compare approaches and learn better implementations.
- Stay consistent: 2–3 hours daily beats 10-hour weekend sessions for retention.